

## CHAPTER 18 Control of Gene Expression

**Concept 18.2** Eukaryotic gene expression is regulated at many stages

Regulation of gene expression is essential for cell specialization in multicellular organisms.

### Differential Gene Expression

- Almost all the cells in a multicellular organism contain an identical genome.
- A typical human cell might express about 20% of its protein-coding genes at any given time.

**House-keeping genes** A subset of genes is expressed in each cell type.

- The differences between cell types, are due not to different genes being present, but to **differential gene expression**. The uniquely expressed genes allow these cells to carry out their specific function. [Click here to view Figure 18.6](#)
- Therefore, the function of any cell, depends on the appropriate set of genes being expressed.

**Differential gene expression** the expression of different genes by cells with the same genome.

- In all organisms, **gene expression** is commonly controlled at transcription, and is often equated with transcription for both bacteria and eukaryotes.
- The greater complexity of eukaryotic cell structure and function provides opportunities for regulating gene expression at many additional stages.

### Regulation of Chromatin Structure

The structural organization of chromatin not only packs a cell's DNA into a compact form that fits inside the nucleus, but also helps regulate gene expression in several ways.

**Heterochromatin** is highly condensed, the genes within it are usually not expressed.

- Certain **chemical modifications to chromatin** can influence chromatin structure and gene expression. (for example, modifications both to the histone proteins of the nucleosomes wrapping the DNA and to the nucleotides that make up that DNA)

### *Histone Modifications and DNA Methylation* [Click here to view Figure 18.7](#)

**Histone tails** are the N-terminus of each histone protein in a nucleosome that protrudes outward from the nucleosome. They are accessible to various modifying enzymes that catalyze the addition or removal of specific chemical groups. (Such as, acetyl, methyl, and phosphate groups).

**Histone acetylation** the addition of an acetyl group to an amino acid in a histone tail. Appears to promote transcription by opening up the chromatin structure.

**DNA methylation** can lead to the condensation of chromatin and reduced transcription. The DNA of individual genes is usually more heavily methylated in cells in which those genes are not expressed. Removal of the extra methyl groups can turn on some of these genes.

- A different set of enzymes can methylate the DNA itself on certain bases, usually cytosine.
- DNA methylation occurs in most plants, animals, and fungi.
- Long stretches of inactive DNA—such as that of inactivated mammalian X chromosomes—are generally more methylated than regions of actively transcribed DNA.
- Once methylated, genes usually stay that way through successive cell divisions in a given individual. Thus, methylation patterns are passed on to daughter cells.
- **Genomic imprinting** in mammals, where methylation permanently regulates expression of either the maternal or paternal allele of particular genes at the start of development

### ***Epigenetic Inheritance***

**Epigenetic inheritance** inheritance of traits transmitted by mechanisms not involving the nucleotide sequence itself. EXAMPLE the chromatin modifications do not change the DNA sequence, yet they still may be passed along to future generations of cells.

### **NOTES**

- Mutations in the DNA are permanent changes, modifications to the chromatin can be reversed. EXAMPLE DNA methylation patterns are largely erased and reestablished during gamete formation.
- Alterations in normal patterns of DNA methylation are seen in some cancers, where they are associated with inappropriate gene expression.

### **Regulation of Transcription Initiation**

Chromatin-modifying enzymes **provide** initial control of gene expression **by** making a region of DNA either more or less able to bind the transcription machinery.

- The regulation of transcription initiation in eukaryotes involves proteins that bind to DNA and either facilitate or inhibit binding of RNA polymerase.

## Organization of a Typical Eukaryotic Gene and Its Transcript

**RECALL** A cluster of proteins called a *transcription initiation complex* assembles on the promoter sequence at the “upstream” end of the gene. One of these proteins, RNA polymerase II, then proceeds to transcribe the gene, synthesizing a primary RNA transcript (pre- mRNA). RNA processing includes enzymatic addition of a 5' cap and a poly-A tail, as well as splicing out of introns, to yield a mature mRNA. [Click here to view Figure 18.8](#)

**Control elements**—associated with most eukaryotic genes—are segments of noncoding DNA that serve as binding sites for the proteins called transcription factors to regulate transcription.

### Types of control elements:

1. **Proximal control elements** are some control elements located close to the promoter.
2. **Distal control elements (Enhancers)** are the more distant control elements from the promoter. May be thousands of nucleotides upstream or downstream of a gene or even within an intron. Each enhancer, is generally associated with only that gene and no other.

## The Roles of General and Specific Transcription Factors

**General Transcription Factors at the Promoter** essential for the transcription of *all* protein-coding genes. A few general transcription factors bind to a DNA sequence such as the TATA box within the promoter, but many bind to proteins, including other transcription factors and RNA polymerase II.

- The interaction of general transcription factors and RNA polymerase II with a promoter usually leads to a **low rate of initiation and production of few RNA transcripts** from genes that are not expressed all the time, but instead are regulated.

### Enhancers and Specific Transcription Factors

**Specific transcription factors** a set of proteins that provide high levels of transcription of genes at the appropriate time and place when control elements interact with it. Either **activators**, which strongly increase gene expression, or **repressors**, which strongly decrease gene expression.

### Types of structural domains found activator proteins:

1. **DNA-binding domain** a part of the protein's three-dimensional structure that binds to DNA.
2. **One or more Activation domains** bind other regulatory proteins or components of the transcription machinery, facilitating a series of protein-protein interactions that result in enhanced transcription of a given gene.

**How can binding of activators to an enhancer located far from the promoter influence transcription?** protein- mediated bending of the DNA is thought to bring the bound activators into contact with a group of *mediator proteins*, which in turn interact with general transcription factors at the promoter. [Click here to view Figure 18.10](#)

- **Protein-protein interactions help** (1) assemble and position the initiation complex on the promoter, and (2) allow the promoter and enhancer to come together in a very specific fashion.

**Different ways that repressors can inhibit gene expression:**

- (1) Binding directly to control element DNA, blocking activator binding.
- (2) Interfering with the activator itself so it can't bind the DNA.

**Some activators and repressors act indirectly by affecting chromatin structure:**

- (1) Some **activators** recruit proteins that **acetylate histones** near the promoters of specific genes, thus **promoting transcription**.
- (2) Some **repressors** recruit proteins that **remove acetyl groups** from histones, leading to **reduced transcription**, a phenomenon referred to as **silencing**.

### **Combinatorial Control of Gene Activation**

- Control of transcription **depends** largely on the binding of activators to DNA control elements.
- The number of completely **different nucleotide sequences** found in control elements is **surprisingly small**. A dozen or so short nucleotide sequences appear again and again in the control elements for different genes.
- Each enhancer is composed of about ten control elements, each of which can bind only one or two specific transcription factors.
- It is the particular **combination of control elements in an enhancer**, rather than the presence of a single unique control element, that **is important in regulating transcription** of the gene.
- Even with only a dozen control element sequences available, a very large number of combinations is possible. The use of **different combinations** of just a few control elements can **allow differential regulation of transcription** in different cell types **because each cell type contains a different group of activator proteins**. [Click here to view Figure 18.11](#)

## *Coordinately Controlled Genes in Eukaryotes*

**Coordinately controlled genes** clustered into an operon, which is regulated by a single promoter and transcribed into a single mRNA molecule. Thus, the genes are expressed together.

- With a few exceptions, operons that work in this way have *not* been found in eukaryotic cells.
- Eukaryotic genes that are co-expressed are typically scattered over different chromosomes.

**Coordinate gene expression depends on** every gene of a dispersed group having a specific combination of control elements. Activator proteins in the nucleus that recognize the control elements bind to them, promoting transcription of the genes, no matter where they are in the genome.

**Coordinate control of dispersed genes in a eukaryotic cell often occurs in response to chemical signals from outside the cell** **EXAMPLE:**

- A **steroid hormone**, enters a cell and binds to a specific intracellular receptor protein, forming a hormone-receptor complex that **serves as a transcription activator**. Every gene whose transcription is stimulated by a given steroid hormone, regardless of its chromosomal location, has a control element recognized by that hormone-receptor complex.
- Many signaling molecules, such as **nonsteroid hormones and growth factors**, bind to receptors on a cell's surface and **never actually enter the cell**. Such molecules can **control gene expression indirectly by triggering signal transduction pathways** that activate particular transcription factors.

**NOTE** Genes with the same sets of control elements are activated by the same chemical signals.

## *Nuclear Architecture and Gene Expression*

**Chromosome conformation capture (3C) techniques** allow researchers to cross-link and identify regions of chromosomes that associate with each other during interphase.

**These studies revealed** that loops of chromatin extend from individual chromosomal territories into specific sites in the nucleus. Different loops from the same chromosome and loops from other chromosomes may congregate in such sites forming a **transcription factory**, some of which are rich in RNA polymerases and other transcription-associated proteins.

**Transcription factories** areas specialized for a common function. [Click here to view Figure 18.12](#)

**NOTE** Genes that are not being expressed are located in the outer edges of the nucleus, while those that are being expressed are found in its interior region.

## Mechanisms of Post-Transcriptional Regulation

### *RNA Processing*

**Alternative RNA splicing** regulation at the RNA-processing level, in which different mRNA molecules are produced from the same primary transcript, depending on which RNA segments are treated as exons and which as introns. [Click here to view Figure 18.13](#)

- Regulatory proteins specific to a cell type **control** intron-exon choices **by** binding to regulatory sequences within the primary transcript.

**Alternative splicing** was proposed as one **explanation for the surprisingly low number of human genes** counted when the human genome was sequenced. EXPLAINED:

- **What, if not the number of genes, accounts for the more complex morphology (external form) of humans?** It turns out that more than **90%** of human protein-coding genes probably undergo alternative splicing. Thus, the extent of **alternative splicing greatly multiplies the number of possible human proteins**, which may be better correlated with complexity of form.

### *Initiation of Translation and mRNA Degradation*

For some mRNAs, the initiation of translation can be blocked by regulatory proteins that bind to specific sequences or structures within the untranslated region (UTR) at the 5' or 3' end, preventing the attachment of ribosomes.

**In a eukaryotic cell**, such “global” control usually involves the activation or inactivation of one or more of the protein factors required to initiate translation.

- EXAMPLE mRNAs that are stored in eggs, just after fertilization, translation is triggered by the sudden activation of translation initiation factors. The response is a burst of synthesis of the proteins encoded by the stored mRNAs.

**Plants and algae** store mRNAs during periods of darkness; light then triggers the reactivation of the translational apparatus.

### **The life span of mRNA:**

- (1) **Bacterial mRNA** molecules typically are degraded by enzymes within a few minutes of their synthesis. This **short life span** of mRNAs is one reason bacteria can change their patterns of protein synthesis so quickly in response to environmental changes.
- (2) **Multicellular eukaryotes' mRNAs** typically survive for hours, days, or even weeks. **For instance**, the mRNAs for the hemoglobin polypeptides, are unusually stable, and these long-lived mRNAs are translated repeatedly in red blood cells.

**NOTE** Nucleotide sequences that affect how long an mRNA remains intact are **often found in** the untranslated region at the 3' end of the molecule.

### ***Protein Processing and Degradation***

Often, eukaryotic polypeptides must be processed to yield functional protein molecules. **For instance**, cleavage of the initial insulin polypeptide (pro-insulin) forms the active hormone.

- Regulatory proteins are commonly activated or inactivated **by** the reversible addition of **phosphate groups**.
- Proteins destined for the surface of animal cells acquire sugars.
- Cell-surface proteins and many others must also be transported to target destinations in the cell in order to function.

The length of **time each protein functions** in the cell is strictly **regulated by selective degradation**. For instance, **cyclins** involved in regulating the cell cycle, **must be relatively short-lived** if the cell is to function appropriately.

#### **To mark a protein for destruction:**

1. The cell commonly **attaches** molecules of a small protein called **ubiquitin to the protein**.
2. Giant protein complexes called **Proteasomes** recognize the ubiquitin-tagged proteins and degrade them.



### **Concept 18.3** Noncoding RNAs play multiple roles in controlling gene expression

- Genome sequencing has revealed that protein-coding DNA accounts for only 1.5% of the human genome. A very small fraction of the non-protein-coding DNA consists of genes for RNAs such as ribosomal RNA and transfer RNA.
- A study of the entire human genome showed that roughly 75% of the genome is transcribed at some point in any given cell.
- Introns account for only a fraction of this transcribed, nontranslated RNA.
- These results suggest that a significant amount of the genome may be transcribed into **non-protein-coding RNAs (also called *noncoding RNAs*, or *ncRNAs*)** including a variety of small RNAs.

### **Effects on mRNAs by MicroRNAs and Small Interfering RNAs**

#### **(1) MicroRNAs (miRNAs)**

- A longer RNA precursor is processed by cellular enzymes into **small, single-stranded RNA** molecules capable of binding to complementary sequences in mRNA molecules.
- miRNAs are **about 22 nucleotides** that forms a complex with one or more proteins. The miRNA allows the complex to bind to any mRNA molecule with at least 7 or 8 nucleotides of complementary sequence.
- The miRNA-protein complex then **degrades the target mRNA or, less often, simply blocks its translation**. [Click here to view Figure 18.14](#)

NOTE Biologists estimate that expression of at least one-half of all human genes may be regulated by miRNAs.

#### **(2) Small interfering RNAs (siRNAs)** similar in size and function to miRNAs

- If siRNA precursor RNA molecules are injected into a cell, the cell's machinery can process them into siRNAs that turn off expression of genes
- **The distinction between miRNAs and siRNAs** is based on subtle differences in the structure of their precursors, which in both cases are RNA molecules that are mostly double-stranded.



**RNA interference (RNAi)** the blocking of gene expression by siRNAs. It is **used for** disabling specific genes to investigate their function.

- **RNAi pathway can process double-stranded RNAs** into homing devices that lead to **destruction of related RNAs**, some scientists think that this pathway may have evolved as a natural defense against infection by such viruses.
- RNAi can also affect the expression of nonviral cellular genes may reflect a different evolutionary origin for the RNAi pathway.

NOTE many species, including mammals, produce their own long, double-stranded RNA precursors to small RNAs such as siRNAs which then can interfere with gene expression at stages other than translation.

### **Chromatin Remodeling and Effects on Transcription by ncRNAs**

Some **ncRNAs** act to bring about **remodeling of chromatin structure**. (for instance, during formation of heterochromatin at the centromere)

- In the **S phase**, the centromeric regions of DNA must be loosened for chromosomal replication and then re-condensed into heterochromatin in preparation for mitosis.
- siRNAs produced by the yeast cells themselves are required to condense the centromere chromatin into heterochromatin by interacting with other ncRNAs and with chromatin-modifying enzymes. [Click here to view Figure 18.15](#)
- In mammalian cells, siRNAs are not known to occur.

#### **(1) Piwi-interacting RNAs, or piRNAs**

- They **induce formation of heterochromatin, blocking expression of** some parasitic DNA elements in the genome known as **transposons**.
- Usually **24–31 nucleotides** in length, piRNAs are processed from a longer, single-stranded RNA precursor.
- They appear to help **reestablish appropriate methylation patterns** in the genome **during gamete formation**.

## (2) Long noncoding RNAs (lncRNAs)

- Ranging from 200 to hundreds of thousands of nucleotides in length

### Roles of Long noncoding RNAs (lncRNAs)

1. **Responsible for X chromosome inactivation**, which, in most female mammals, **prevents expression of genes located on one of the X chromosomes**. EXPLAINED lncRNAs bind back to and coat that chromosome, and this binding **leads to condensation of the entire chromosome into heterochromatin**.
2. Act as a scaffold, **bringing together DNA, proteins, and other RNAs into complexes**. These associations may act **either (1)** to condense chromatin **or (2)** to help bring the enhancer of a gene together with mediator proteins and the gene's promoter, activating gene expression in a more direct fashion.

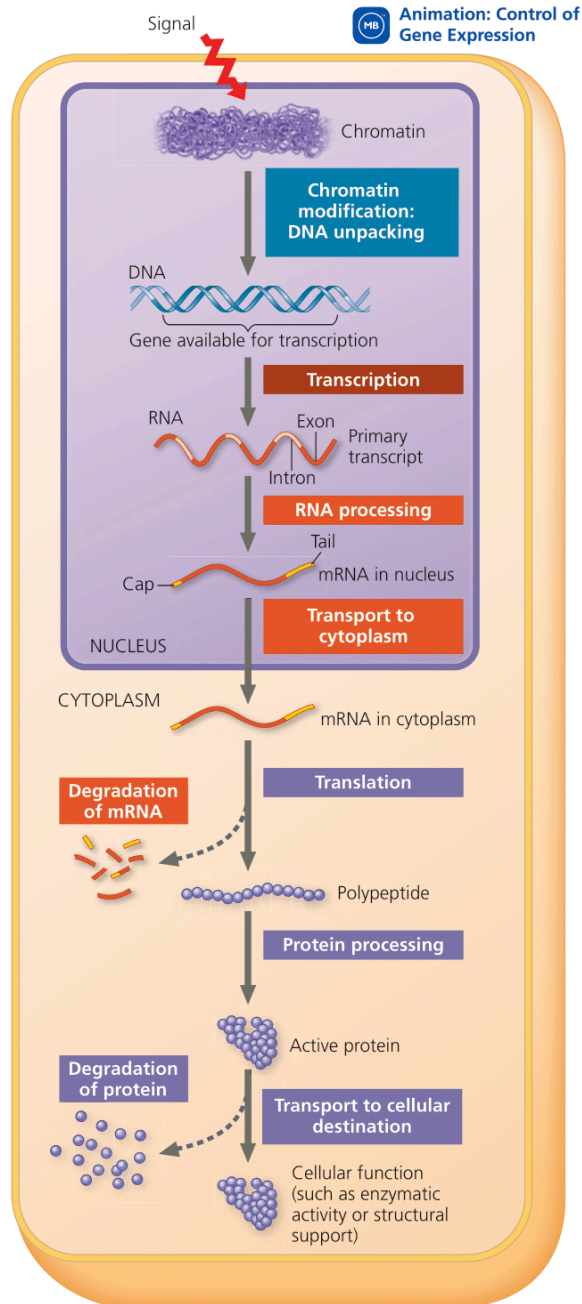
## The Evolutionary Significance of Small ncRNAs

### EVOLUTION

- Small ncRNAs are also used by bacteria as a defense system, called the CRISPR-Cas9 system, against viruses that infect them.
- The versatility of miRNA regulation has therefore led some biologists to hypothesize that an increase in the number of different miRNAs specified by the genome of a given species has allowed morphological complexity to increase over evolutionary time.
- A survey of different species supports the notion that siRNAs evolved first, followed by miRNAs and later piRNAs, which are found only in animals.

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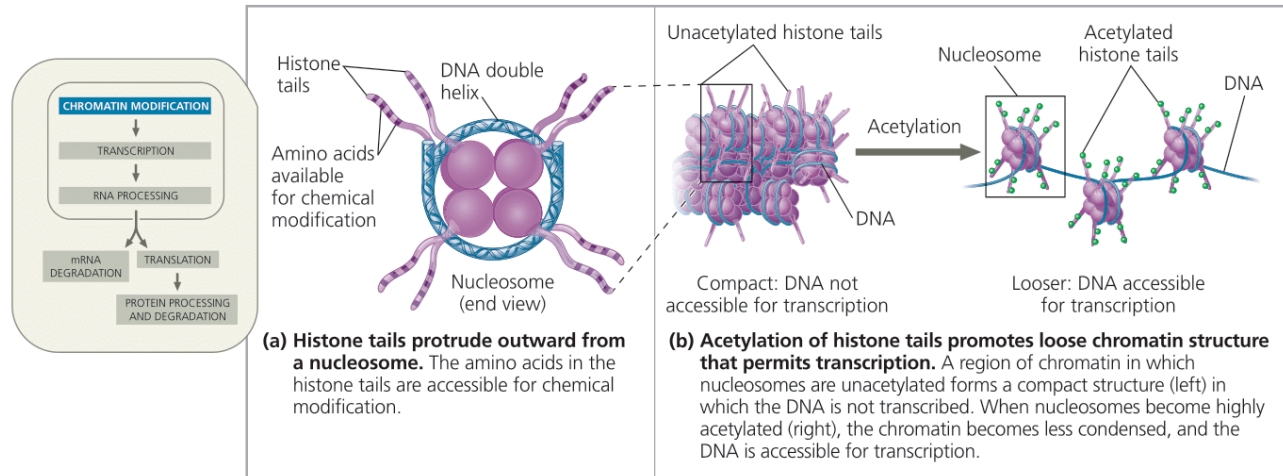
**▼ Figure 18.6 Stages in gene expression that can be regulated in eukaryotic cells.** In this diagram, the colored boxes indicate the processes most often regulated; each color indicates the type of molecule that is affected (blue = DNA, red/orange = RNA, purple = protein). The nuclear envelope separating transcription from translation in eukaryotic cells offers an opportunity for post-transcriptional control in the form of RNA processing that is absent in prokaryotes. In addition, eukaryotes have a greater variety of control mechanisms operating before transcription and after translation. A miniature version of this figure accompanies several figures later in the chapter as an orientation diagram.



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### ▼ Figure 18.7 A simple model of histone tails and the effect of histone acetylation.

In addition to acetylation, histones can undergo several other types of modifications, such as methylation or phosphorylation. These also help determine the chromatin configuration in a region, sometimes by establishing binding sites for chromatin-modifying enzymes.



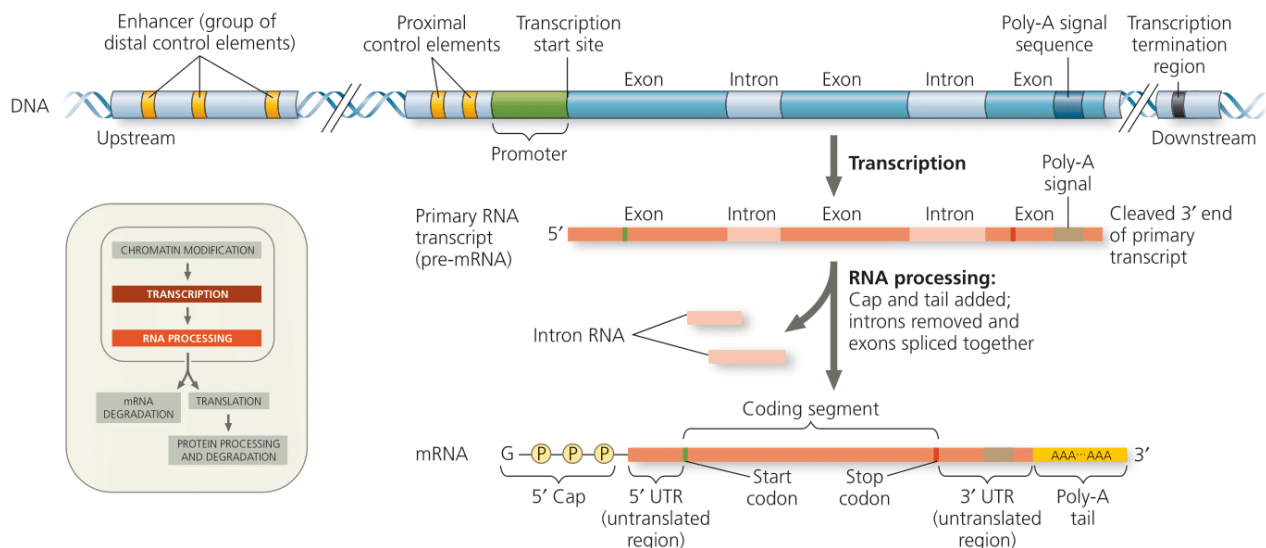
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### ▼ Figure 18.8 A eukaryotic gene and its transcript.

Each eukaryotic gene has a promoter—a DNA sequence where RNA polymerase binds and starts transcription, proceeding “downstream.” A number of control elements (gold) are involved in regulating the initiation of transcription; these are DNA sequences located near (proximal to) or far from

(distal to) the promoter. Distal control elements can be grouped together as enhancers, one of which is shown for this gene. At the other end of the gene, a polyadenylation (poly-A) signal sequence in the last exon of the gene is transcribed into an RNA sequence that signals where the transcript is cleaved and the poly-A tail added. Transcription may continue for hundreds

of nucleotides beyond the poly-A signal before terminating. RNA processing of the primary transcript into a functional mRNA involves three steps: addition of the 5' cap, addition of the poly-A tail, and splicing. In the cell, the 5' cap is added soon after transcription is initiated, and splicing occurs while transcription is still under way (see Figure 17.11).

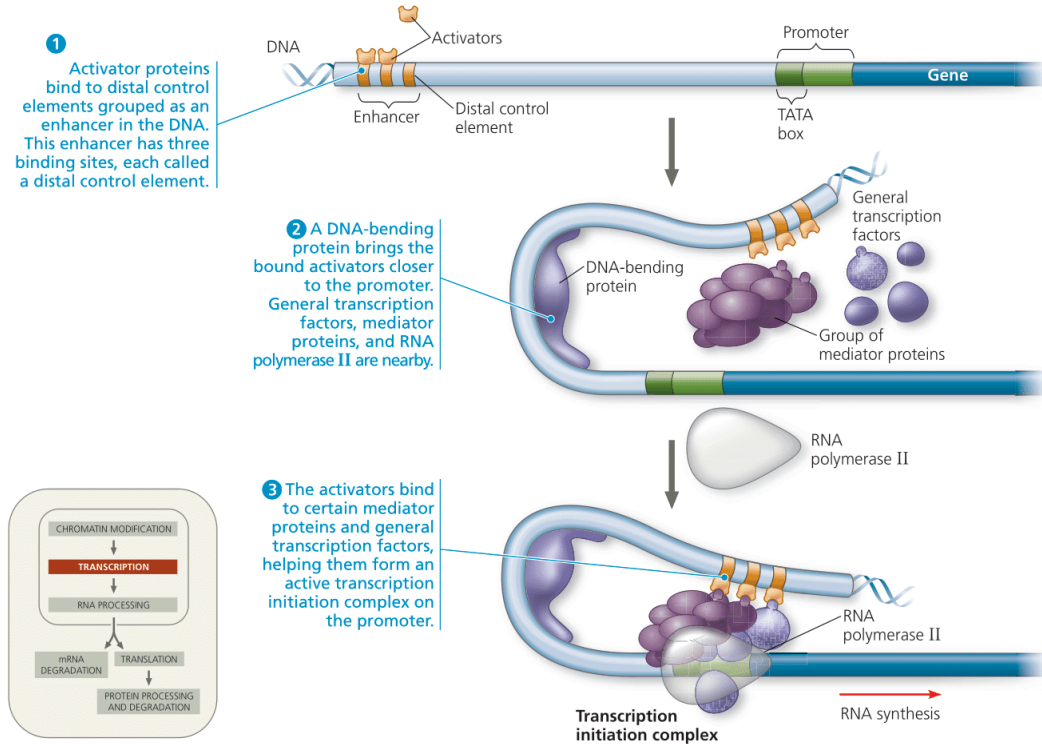


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▼ **Figure 18.10 A model for the action of enhancers and transcription activators.** Bending of the DNA by a protein enables enhancers to influence a promoter hundreds or even thousands of nucleotides away. Specific transcription factors called

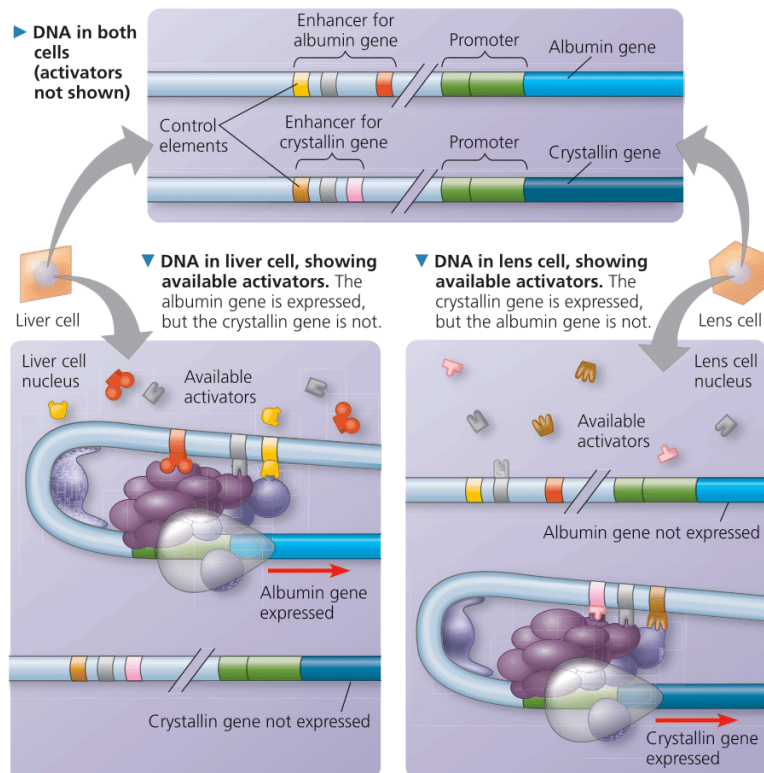
activators bind to the enhancer DNA sequences and then to a group of mediator proteins. These in turn bind to general transcription factors and then RNA polymerase II, thus assembling the transcription initiation complex. These protein-protein interactions lead to

correct positioning of the complex on the promoter and the initiation of RNA synthesis. Only one enhancer (with three gold control elements) is shown here, but a gene may have several enhancers that act at different times or in different cell types.



► **Figure 18.11 Cell type-specific transcription.** Both liver cells and lens cells have the genes for making the proteins albumin and crystallin, but only liver cells make albumin (a blood protein) and only lens cells make crystallin (the main protein of the lens of the eye). The specific transcription factors made in a cell determine which genes are expressed. In this example, the genes for albumin and crystallin are shown at the top, each with an enhancer made up of three different control elements. Although the enhancers for the two genes both have a gray control element, each enhancer has a unique combination of elements. All the activator proteins required for high-level expression of the albumin gene are present in liver cells only (left), whereas the activators needed for expression of the crystallin gene are present in lens cells only (right). For simplicity, we consider only the role of specific transcription factors that are activators here, although repressors may also influence transcription in certain cell types.

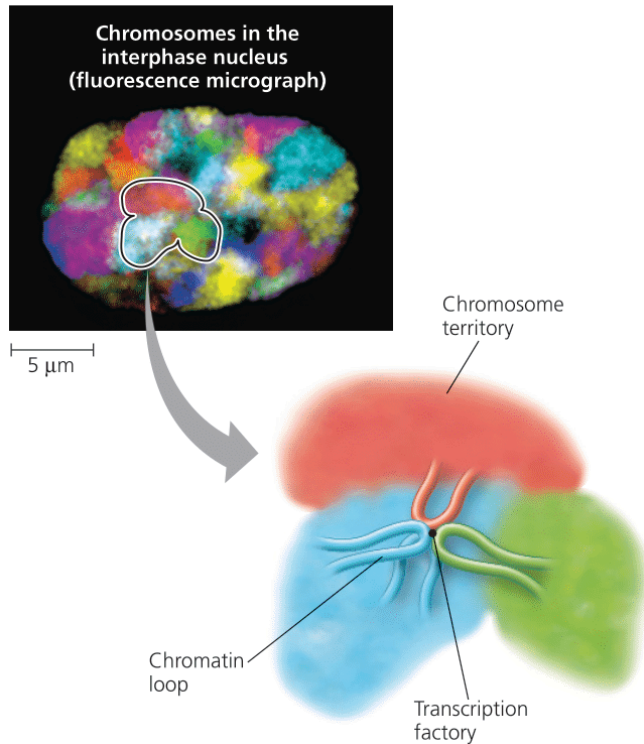
**VISUAL SKILLS** ► Describe the enhancer for the albumin gene in each type of cell. How would the nucleotide sequence of this enhancer in the liver cell compare with that in the lens cell?





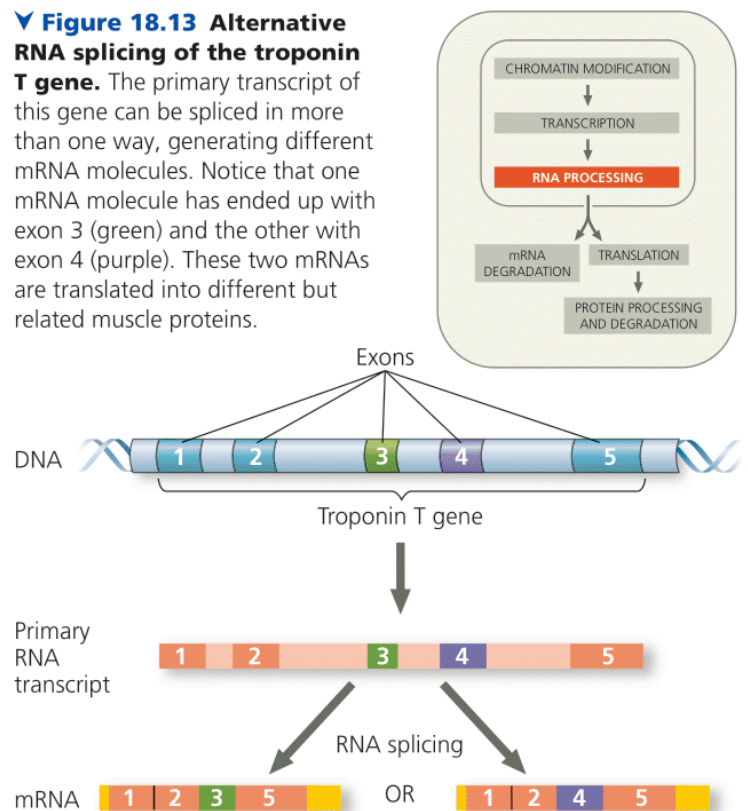
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▼ **Figure 18.12 Chromosomal interactions in the interphase nucleus.** Although each chromosome has its own territory (see Figure 16.23b), loops of chromatin may extend into other sites in the nucleus. Some of these sites are transcription factories that are occupied by multiple chromatin loops from the same chromosome (blue loops) or other chromosomes (red and green loops).

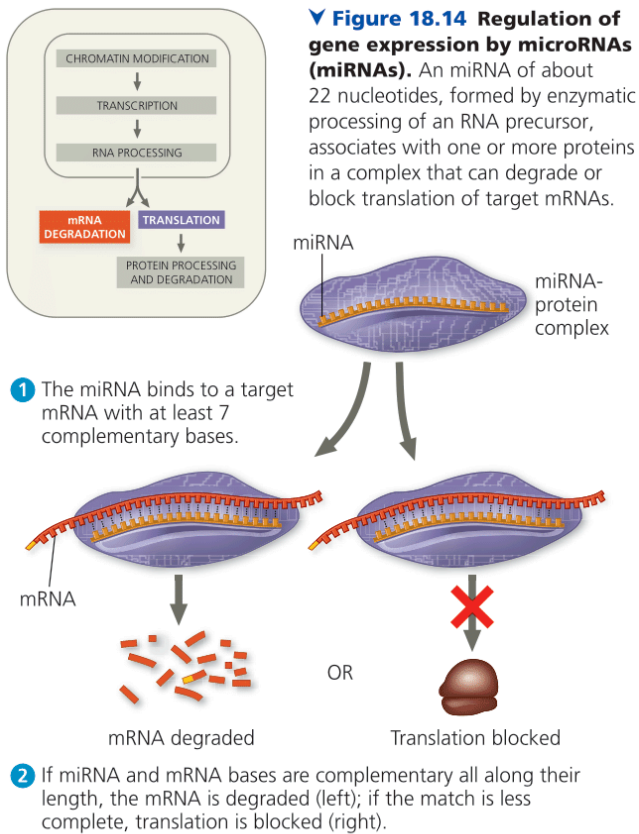


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▼ **Figure 18.13 Alternative RNA splicing of the troponin T gene.** The primary transcript of this gene can be spliced in more than one way, generating different mRNA molecules. Notice that one mRNA molecule has ended up with exon 3 (green) and the other with exon 4 (purple). These two mRNAs are translated into different but related muscle proteins.



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**▼ Figure 18.15 Condensation of chromatin at the centromere.**

In one type of yeast, siRNAs and longer noncoding RNAs cooperate in the pathway that leads to re-formation of highly condensed heterochromatin at the centromere of each chromatid after DNA replication.

